

Functions

A function is a relation which assigns each element $x \in X$ to exactly one element $y \in Y$.

"Input" : Domain

"Output" : Range

Test for functions : Vertical line test

Inverse f^{-1} $ff^{-1}(x) = \emptyset$ to

For inverse function f^{-1} to exist, function f must be an one-one

Test for one-one functions : Horizontal line test

The line $y = k$ intersects the graph at most once.

$\therefore$ Function is one-one

$\therefore f^{-1}$ exists.

$$(fg)^{-1} = g^{-1}f^{-1}$$

To define f^{-1} : Let $y = f(x)$ then make x the subject.

$$D_{f^{-1}} = R_f$$

$$R_{f^{-1}} = D_f$$

(: Proper subset

[: Including

Composite gf

For composite function gf to exist, $R_f \subseteq D_g$

$$D_{gf} = D_f$$

$$R_{gf} \subseteq R_g$$

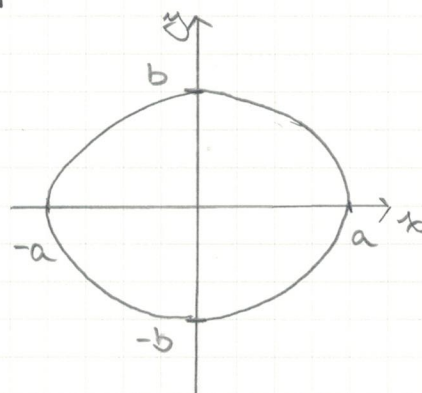
To find D_{gf} , put $f(x)$ into $g(x)$.

Graphs and Transformations

Common graphs:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

→ Ellipse or circle if $a = b$

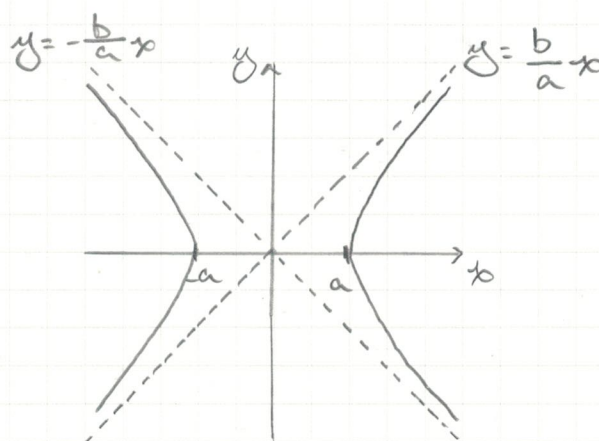


$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

→ Hyperbola

→ Asymptotes: $y = \frac{b}{a}x$

$$y = -\frac{b}{a}x$$

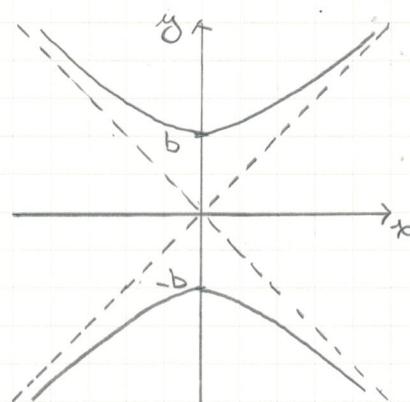


$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$$

→ Hyperbola

→ Asymptotes: $y = \frac{b}{a}x$

$$y = -\frac{b}{a}x$$



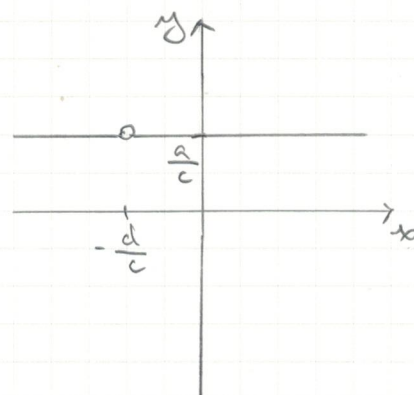
$$y = \frac{ax + b}{cx + d}$$

(Rational functions)

→ If $(cx + d)$ is a factor of $(ax + b)$

→ Horizontal line of $y = \frac{a}{c} = \frac{b}{d}$

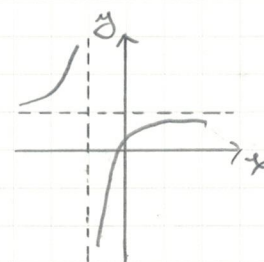
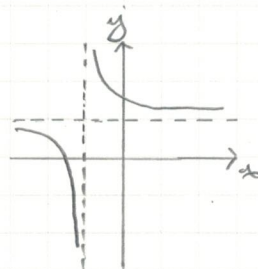
→ Undefined at $x = -\frac{d}{c}$



→ If $\frac{a}{c} \neq \frac{b}{d}$

→ Asymptotes: $y = \frac{a}{c}$

$$x = -\frac{d}{c}$$

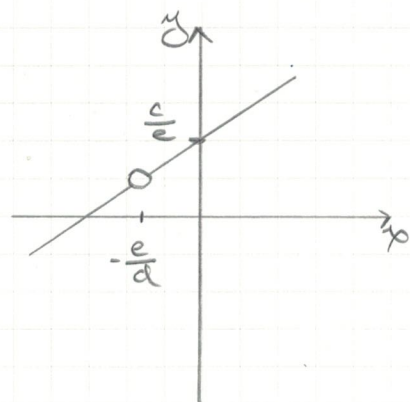


$$y = \frac{ax^2 + bx + c}{dx + e}$$

→ If $(dx + e)$ is a factor of $(ax^2 + bx + c)$

→ Straight line of $y = \frac{a}{d}x + \frac{c}{e}$

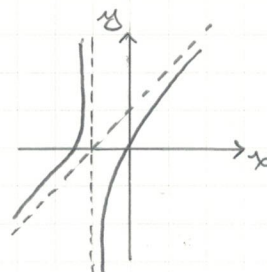
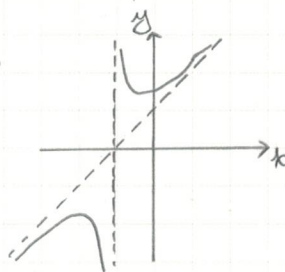
→ Undefined at $x = -\frac{e}{d}$



→ If $(dx + e)$ is not a factor of $(ax^2 + bx + c)$

→ Asymptotes: $y = \frac{a}{d}x + k$,

$$x = -\frac{e}{d}$$



Asymptotes

Vertical asymptotes: $y \rightarrow \pm\infty$ as $x \rightarrow k$

→ For rational functions, $f(x) = \frac{p(x)}{q(x)}$

asymptote can be found by equating $q(x) = 0$

Horizontal asymptotes: $y \rightarrow k$ as $x \rightarrow \infty$

→ For rational functions, there is at most one horizontal asymptote.

→ If degree of $p(x) <$ degree of $q(x)$, asymptote is $y = 0$.

→ If degree of $p(x) =$ degree of $q(x)$, asymptote is $y = \frac{a}{c}$

Oblique asymptote

→ For rational functions, if degree of $p(x) >$ degree of $q(x)$,

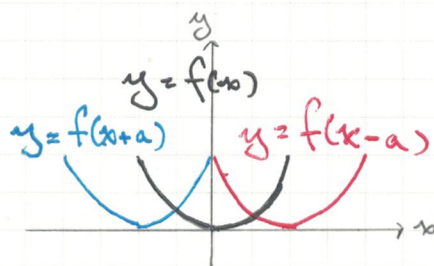
asymptote is non-remainder part of $\frac{p(x)}{q(x)}$.

Transformation

Translate

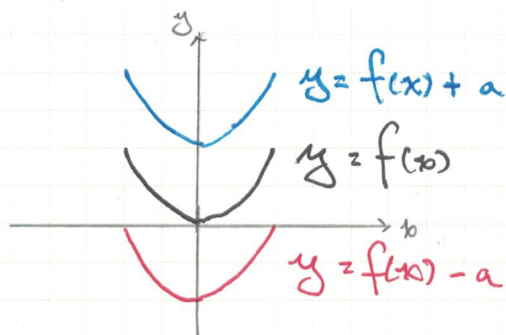
$$y = f(x) \rightarrow y = f(x-a)$$

Translate graph of $y = f(x)$ in the positive x -axis by a units.



$$y = f(x) \rightarrow y = f(x) + a$$

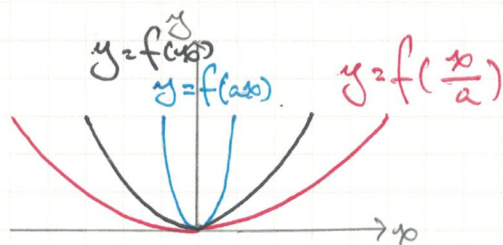
Translate graph of $y = f(x)$ in the positive y -axis by a units.



Scale

$$y = f(x) \rightarrow y = f(ax)$$

Scale graph of $y = f(x)$ in the x -axis by factor $1/a$.



$$y = f(x) \rightarrow y = af(x)$$

Scale graph of $y = f(x)$ in the y -axis by factor a .



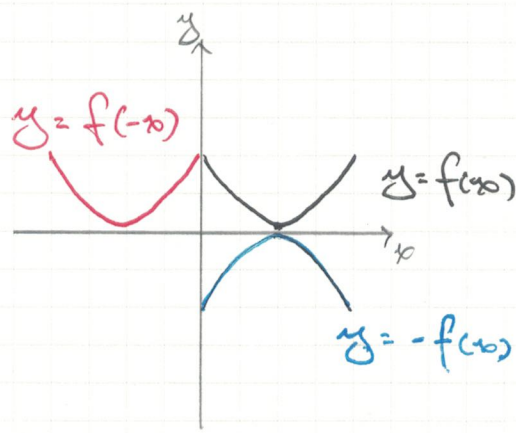
Reflect

$$y = f(x) \rightarrow y = -f(x)$$

Reflect graph of $y = f(x)$ about the x -axis.

$$y = f(x) \rightarrow y = f(-x)$$

Reflect graph of $y = f(x)$ about the y -axis.



Sequences and Series

Sequence : A ordered list with a rule relating the value of each term to its position in the list.

Series : Summation of all the terms in a sequence.

* Series is convergent if $S_n \rightarrow k$ as $n \rightarrow \infty$

Properties

$$\sum_{r=i}^n [a u_r + b v_r] = a \sum_{r=i}^n u_r + b \sum_{r=i}^n v_r$$

* $\lim_{n \rightarrow \infty} (S_n)$: Limit of S_n when n approaches ∞ .

$$\sum_{r=i}^n u_r = \sum_{r=i}^{k-1} u_r + \sum_{r=k}^n u_r \quad (i < k < n)$$

* Number of terms stay constant

Standard series

$$\sum_{r=1}^n r = \frac{1}{2} n(n+1)$$

$$\sum_{r=1}^n r^2 = \frac{1}{6} n(n+1)(2n+1)$$

$$\sum_{r=1}^n r^3 = \frac{1}{4} n^2(n+1)^2$$

* $u_n = S_n - S_{n-1}$

⇒ If u_n is a polynomial in n of degree k , S_n is a polynomial in n of degree $k+1$.

Arithmetic progression : Sequence where the difference between any term and its immediate previous term is a constant called the common difference.

$$u_n = a + (n-1)d$$

$$S_n = \frac{1}{2} n [2a + (n-1)d] = \frac{1}{2} n (a + u_n)$$

Test for A.P.

- $u_{n+1} - u_n = \text{constant}$ (Independent of n)

- $u_n = a + (n-1)d$

- x, y and z are in A.P. $\Leftrightarrow x + z = 2y$

Geometric progression : Sequence where the ratio between any term and its immediate previous term is a constant called the common ratio.

$$u_n = ar^{n-1}$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}$$

Test for G.P.

$$- \frac{u_{n+1}}{u_n} = \text{Constant (Independent of } n)$$

$$- u_n = a r^{n-1}$$

$$- x, y \text{ and } z \text{ are in G.P.} \Leftrightarrow xy = y^2$$

$$S_{\infty} = \frac{a}{1-r}$$

* Geometric series is convergent if $|r| < 1$ and $S_n \rightarrow k$ as $n \rightarrow \infty$

$$* S_n = \frac{a(1-r^n)}{1-r} = S_{\infty}(1-r^n)$$

Differentiation

Rules

$$\frac{d}{dx}(uv) = uv' + vu'$$

$$\frac{d}{dx} \frac{u}{v} = \frac{u'v - v'u}{v^2}$$

Parametric:

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$$

$$= \frac{d}{dx} \frac{d}{dt} \left(\frac{dy}{dx} \right) \times \frac{dt}{dx}$$

$$f(x)$$

$$\sin f(x)$$

$$\cos f(x)$$

$$\tan f(x)$$

$$\cot f(x)$$

$$f'(x)$$

$$f'(x) \cos f(x)$$

$$f'(x) \sin f(x)$$

$$f'(x) \sec^2 f(x)$$

$$f'(x) \operatorname{cosec}^2 f(x)$$

$$af(x)$$

$$af(x) \Rightarrow a f'(x)$$

Integration

Techniques

Polynomial:

$$\boxed{n \neq -1} \int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

$$\boxed{n = -1} \int f'(x) [f(x)]^{-1} dx = \int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

Exponential and Logarithmic:

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c, \quad a > 0$$

Trigonometric:

$$\int f'(x) \cos[f(x)] dx = \sin[f(x)] + c$$

$$\int f'(x) \sin[f(x)] dx = -\cos[f(x)] + c$$

$$\int f'(x) \sec^2[f(x)] dx = \tan[f(x)] + c$$

$$\int f'(x) \sec[f(x)] \tan[f(x)] dx = \sec[f(x)] + c$$

$$\int f'(x) \operatorname{cosec}[f(x)] \cot[f(x)] dx = -\operatorname{cosec}[f(x)] + c$$

$$\int f'(x) \operatorname{cosec}^2[f(x)] dx = -\cot[f(x)] + c$$

Inverse:

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \frac{1}{a} \sin^{-1} \left[\frac{f(x)}{a} \right] + c$$

$$\int \frac{f'(x)}{[f(x)]^2 + a^2} dx = \frac{1}{a} \tan^{-1} \left[\frac{f(x)}{a} \right] + c$$

$$\int \frac{f'(x)}{[f(x)]^2 - a^2} dx = \frac{1}{2a} \ln \left[\frac{f(x) - a}{f(x) + a} \right] + c$$

$$\int \frac{f'(x)}{a^2 - [f(x)]^2} dx = \frac{1}{2a} \ln \left[\frac{a + f(x)}{a - f(x)} \right] + c$$

* Factor formulae

$$\int \sin(mx) \cos(nx) dx$$

$$\int \cos(mx) \cos(nx) dx$$

$$\int \sin(mx) \sin(nx) dx$$

* Double angle formulae

$$\int \cos^2(x) dx = \frac{1}{2} \int (1 + \cos(2x)) dx$$

$$\int \sin^2(x) dx = \frac{1}{2} \int (1 - \cos(2x)) dx$$

$$\int \tan^2(x) dx = \int \sec^2 x - 1 dx$$

* Partial fractions

* Complete the square

Substitution: $\int f(u) g'(x) dx = \int f(u) du \quad | \quad u = g(x)$

Integration by parts: $\int u dv = uv - \int v du$

Definite integral: $\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$

Area bounded by curve and x-axis: $\int_a^b y dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n A_i$

Area bounded by curve and y-axis: $\int_c^d x dy$



* Area = $\int y_2 - y_1 dx$

Volume of revolution about the x-axis: $\pi \int_a^b y^2 dx$

Volume of revolution about the y-axis: $\pi \int_c^d x^2 dy$

Complex Numbers

$$i = \sqrt{-1}$$

Cartesian form: $z = x + iy$, $x, y \in \mathbb{R}$

$$\operatorname{Re}(z) = x \quad \operatorname{Im}(z) = y$$

Complex numbers z and w are equal if and only if
 $\operatorname{Re}(z) = \operatorname{Re}(w)$ and $\operatorname{Im}(z) = \operatorname{Im}(w)$



$$\text{Complex conjugate } z^* = x - iy = \frac{1}{z}$$

$$\text{Properties: } (z^*)^* = z$$

$$z + z^* = 2\operatorname{Re}(z)$$

$$z - z^* = 2i\operatorname{Im}(z)$$

$$zz^* = \operatorname{Re}(z)^2 + \operatorname{Im}(z)^2$$

$$(z_1 \pm z_2)^* = z_1^* \pm z_2^*$$

Fundamental Theorem of Algebra:

A polynomial equation of degree n has exactly n roots.

Let $f(x)$ be a polynomial with real coefficients only.

If $z = x + yi$ is a root, $z^* = x - yi$ is also a root.

$$\text{Polar form: } z = r(\cos \theta + i \sin \theta)$$

$$r = |z| = \sqrt{x^2 + y^2} \quad * |z| \text{ is the distance from origin to point on the Argand diagram}$$

$$\theta = \arg(z), \quad -\pi < \theta \leq \pi$$

Argument of z is the angle $\vec{OP}$ makes with the positive real axis.

* Principle range: $-\pi < \theta \leq \pi \rightarrow$ Principle argument

$$* \arg(z^*) = -\arg(z)$$

$$\text{Exponential form: } z = re^{i\theta}$$

$$* |z^*| = |z| \quad * |z|^2 = zz^*$$

$$\text{Euler's equation: } e^{i\theta} = \cos \theta + i \sin \theta$$

$$* \arg(iz) = \frac{\pi}{2} + \arg(z)$$

$$* z^* = re^{-i\theta}$$

$$* e^{i\theta} + e^{-i\theta} = 2\cos \theta$$

$$e^{i\theta} - e^{-i\theta} = 2i\sin \theta$$

Vectors I

Vectors are quantities with magnitude and direction.

$$\text{Magnitude of } \underline{a} = |\underline{a}| = \left| \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right| = \sqrt{x^2 + y^2 + z^2}$$

Zero vectors, $\underline{0}$ have zero magnitude

$$\text{Unit vector } \underline{\hat{a}} = \frac{\underline{a}}{|\underline{a}|}$$

$$\text{Vector addition: } \overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$$

* Vector addition is commutative and associative.

Direction cosines are the angles a vector makes with the axes.

$$\overrightarrow{OP} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

Let α, β, γ be the angles $\overrightarrow{OP}$ makes with Ox, Oy, Oz respectively.

$$\cos \alpha = \frac{a}{|\overrightarrow{OP}|} = \frac{a}{\sqrt{a^2 + b^2 + c^2}}$$

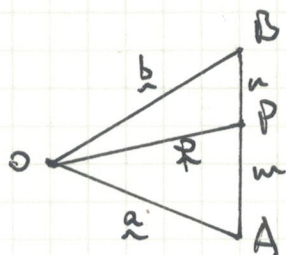
$$\cos \beta = \frac{b}{|\overrightarrow{OP}|} = \frac{b}{\sqrt{a^2 + b^2 + c^2}}$$

$$\cos \gamma = \frac{c}{|\overrightarrow{OP}|} = \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

$\underline{u}$ and $\underline{v}$ are parallel if and only if $\underline{u} = k\underline{v}$, $k \in \mathbb{R} \setminus \{0\}$

* A, B, C are collinear if $\overrightarrow{AB} = k\overrightarrow{BC}$, $k \in \mathbb{R} \setminus \{0\}$

Ratio Theorem:



$$\underline{r} = \frac{1}{m+n} (n\underline{a} + m\underline{b})$$

* if P is midpoint of AB, $\underline{r} = \frac{1}{2}(\underline{a} + \underline{b})$

Vectors II

Scalar Product:

$$\underline{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \quad \underline{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}, \quad \underline{a} \cdot \underline{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta, \text{ where } \theta \text{ is the angle between } \underline{a} \text{ and } \underline{b}$$

* Scalar product is commutative.

$$\underline{a} \cdot \underline{a} = |\underline{a}|^2$$

$$\underline{a} \cdot \underline{b} = 0 \Rightarrow \underline{a} \perp \underline{b}$$

$$\underline{a} \cdot \underline{b} = \underline{a} \cdot \underline{c} \nRightarrow \underline{b} = \underline{c}$$

$$\underline{a} \cdot \underline{b} = \underline{a} \cdot \underline{c} \Rightarrow \underline{a} \cdot \underline{b} - \underline{a} \cdot \underline{c} = 0 \Rightarrow \underline{a} \perp (\underline{b} - \underline{c})$$

$$\text{Angle } \theta \text{ between two vectors} = \cos^{-1} \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| |\underline{b}|} = \cos^{-1}(\underline{\hat{a}} \cdot \underline{\hat{b}})$$

Vector Product:

$$\underline{a} \times \underline{b} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ -(a_1 b_3 - a_3 b_1) \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$$

$$\underline{a} \times \underline{b} = |\underline{a}| |\underline{b}| \sin \theta \underline{\hat{n}}, \text{ where } \theta \text{ is the angle between } \underline{a} \text{ and } \underline{b}$$

$$* \underline{a} \times \underline{b} = -\underline{b} \times \underline{a}$$

* Vector product is not commutative and not associative.

$$\underline{a} \times \underline{b} = 0 \Rightarrow |\underline{a}| = 0 \text{ or } |\underline{b}| = 0 \text{ or } \underline{a} \parallel \underline{b}$$

$$\text{Area of parallelogram} = |\underline{a} \times \underline{b}| \quad \text{Area of triangle} = \frac{1}{2} |\underline{a} \times \underline{b}|$$

$$\text{Projection vector} = |\underline{a}| \cos \theta \underline{\hat{b}} = (\underline{a} \cdot \underline{\hat{b}}) \underline{\hat{b}}, \text{ length} = |\underline{a} \cdot \underline{\hat{b}}|$$

$$\text{Perpendicular distance from A to N on } \overrightarrow{OB} = |\underline{a}| \sin \theta = |\underline{a} \times \underline{\hat{b}}|$$

Vectors III

Vector Equation: $\underline{r} = \underline{a} + \lambda \underline{m}$, where $\lambda \in \mathbb{R}$

$\underline{r}$ gives position vector of any point on the line
 $\underline{a}$ is the position vector of a fixed point on the line
 $\underline{m}$ is the direction vector of the line

Cartesian Equation: $\frac{x-a_1}{m_1} = \frac{y-a_2}{m_2} = \frac{z-a_3}{m_3}$

*Special case: ① $\underline{r} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ m_2 \\ m_3 \end{pmatrix}$, $\mu \in \mathbb{R}$

$\Rightarrow$ line lies in the plane $x = a_1$.

$$\downarrow$$

$$x = a_1, \frac{y-a_2}{m_2} = \frac{z-a_3}{m_3}$$

② $\underline{r} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 0 \\ m_3 \end{pmatrix}$, $\lambda \in \mathbb{R}$

$\Rightarrow$ line lies in the intersection of planes $x = a_1$ and $y = a_2$

$$\downarrow$$

$$x = a_1, y = a_2, z = a_3 + \lambda m_3$$

Relationships between two lines in 3d space:

1. Lines intersect and are coplanar.
2. Lines are skew (non-parallel, non-intersecting)
3. Lines are coincident (parallel and intersecting)
4. Lines are parallel and non-intersecting.

Foot of Perpendicular from a point to a line:

Method 1: Zero Scalar Product

$\Rightarrow$ Given N lies on l, $\vec{ON} = \underline{a} + \lambda \underline{m}$, for some $\lambda \in \mathbb{R}$

$$\Rightarrow \vec{PN} \cdot \underline{m} = 0$$

Length of Perpendicular from a point to a line:

Method 1: Vector Product

$$\Rightarrow |\vec{PN}| = |\vec{AP} \times \hat{\underline{m}}|$$

Coordinates of reflected point:

Method 1: Ratio Theorem

Vectors IV

Vector Equation :

Parametric form : $\underline{r} = \underline{a} + \lambda \underline{b} + \mu \underline{c}$, $\lambda, \mu \in \mathbb{R}$

Scalar product form : $\underline{r} \cdot \underline{n} = p$

OR

$$\underline{r} \cdot \hat{\underline{n}} = p'$$

$\underline{n}$ is the normal vector of plane

p, p' are constants

$|p'|$ is the perpendicular distance of plane from origin $\odot$

Cartesian Equation : $\underline{r} \cdot \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = p \Rightarrow n_1 x + n_2 y + n_3 z = p$

Foot of Perpendicular and Perpendicular distance : To find N, create a line perpendicular to plane $\underline{r} = \underline{q} + \lambda \underline{n}$

3 cases of interaction between line and plane :

$$\ell: \underline{r} = \underline{a} + \lambda \underline{u}$$

$$\pi: \underline{r} \cdot \underline{n} = p$$

① Line is contained in plane
 $\underline{u} \cdot \underline{n} = 0$, $\underline{a} \cdot \underline{n} = p$

② Line is parallel to plane but not contained
 $\underline{u} \cdot \underline{n} = 0$, $\underline{a} \cdot \underline{n} \neq p$

③ Line intersects plane at one point
 $(\underline{a} + \lambda \underline{u}) \cdot \underline{n} = p$

Angle between ℓ and $\pi = \sin^{-1} \left| \frac{\underline{u} \cdot \underline{n}}{|\underline{u}| |\underline{n}|} \right|$, π and $\pi = \cos^{-1} \left| \frac{\underline{u} \cdot \underline{n}}{|\underline{u}| |\underline{n}|} \right|$

Statistics I - Permutation and Combination

Addition Principle: $m + n$

Multiplication Principle: mn

For n distinct objects,
there are $n(n-1)(n-2)\dots(3)(2)(1) = n!$ different permutations.

Given n distinct objects, but we only need to arrange r ,
there are ${}^nP_r = \frac{n!}{(n-r)!}$ different permutations.

Given n objects, of which n_1 are alike, n_2 are alike $\dots$ n_r are alike, there are $\frac{n!}{n_1! n_2! \dots n_r!}$ different permutations.

In a circle, for n distinct objects,
there are $\frac{n!}{n} = (n-1)!$ permutations.

A combination is the selection of objects in which the order of the objects (Specific arrangement) is NOT important.

If r distinct objects are selected from n distinct objects,
the total number of selections is: $\frac{\text{Permutations of } r \text{ objects from } n}{\text{Permutations of } r \text{ objects in each selection}}$

$$= \frac{n!}{r! (n-r)!}$$

$$= \binom{n}{r}$$

Statistics II - Probability

$$P(A) = \frac{\text{Number of ways event A can occur}}{\text{Total number of possible outcome}}$$

Complementary events: $P(A) + P(A') = 1$

Addition Laws:

Principle of Inclusion and Exclusion: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Mutually Exclusive:

If A and B are mutually exclusive events, $P(A \cup B) = P(A) + P(B)$, $P(A \cap B) = 0$

Conditional Probability:

Probability that A will happen given that B has happened =

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Independent Events:

When A and B are independent,

$$P(A|B) = P(A)$$

$$P(B|A) = P(B)$$

$$P(A \cap B) = P(A) \times P(B)$$

For A and B to be exhaustive, $P(A \cup B) = 1$

Statistics III - Discrete Random Variables

Discrete random variable X is a random variable that can only take a finite or countably infinite number of values.

When each value of X has the same probability of occurring, it is a discrete uniform distribution.

Expectation (mean) of X , $E(X) = \sum_{\text{all } x} x P(X=x)$

$E(X)$ may be interpreted as the average value of X in an infinitely large number of trials and is known as mean, μ of X .

$$E(f(x)) = \sum_{\text{all } x} f(x) P(X=x)$$

Properties: $E(f(x) + g(x)) = E(f(x)) + E(g(x))$

$$E(k) = k$$

$$E(kX) = kE(X)$$

Variance (degree of deviation) of X , $\text{Var}(X) = E[(X - \mu)]^2$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

$$= E(X^2) - [E(X)]^2$$

$$\text{Standard deviation } \sigma = \sqrt{\text{Var}(X)}$$

* $[E(X)]^2$ is also written as $E^2(X)$

Independent random variables :

If X and Y are independent random variables,

$$E(aX \pm bY) = aE(X) \pm bE(Y)$$

$$\text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$$

$$* X_1 + X_2 \neq 2X$$

If $X_1, X_2, \dots, X_n$ of the random variable X are independent observations,

$$E(X_1 + X_2 + \dots + X_n) = nE(X) = E(X_1) + E(X_2) + \dots + E(X_n)$$

$$\text{Var}(X_1 + X_2 + \dots + X_n) = n \text{Var}(X) = \text{Var}(X_1) + \dots + \text{Var}(X_n)$$

Statistics IV - Binomial Distribution

Binomial distribution: $P(X=x) = \binom{n}{x} p^x q^{n-x} = \binom{n}{x} p^x (1-p)^{n-x}$

$$X \sim B(n, p), \quad n \in \mathbb{Z}^+, \quad 0 \leq p \leq 1$$

Conditions:

- Trials are independent
- Probability p is constant

$$E(X) = np \quad \text{Var}(X) = npq = np(1-p)$$

$\text{binompdf}(\) \rightarrow P(X=x)$ $\text{binomcdf}(\) \rightarrow P(X \leq x)$

To find:

Mode (Value of X that is the most probable to occur)

$\rightarrow$ Edit ... $\rightarrow \text{binompdf}(\) \rightarrow \text{Sort D}(L_2, L_1)$

List out the values of X one before and after.

X value for $P(X \leq r) > k$ or $< k$

$\rightarrow$ Edit ... $\rightarrow \text{binomcdf}(\) \rightarrow$ Find value of L .

OR

$\rightarrow$ Plot $y_1 = \text{binomcdf} - k$

Using GC,

Find point where $y=0$ in graph / Table

n value or p value for $P(X \leq r) > k$ or $< k$

$\rightarrow$ Plot $y_1 = \text{binomcdf} - k$

Using GC,

Find point where $y=0$ in graph / Table

$$P(X > r) = 1 - P(X \leq r)$$

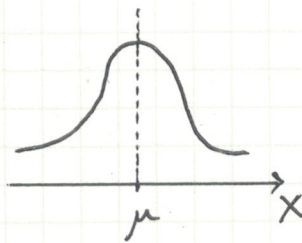
$$P(X \geq r) = 1 - P(X \leq r-1)$$

$$P(X < r) = P(X \leq r-1)$$

Statistics V - Normal Distribution

Normal distribution : $f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{(x-\mu)}{\sigma}\right]^2}$, $\sigma > 0$

$X \sim N(\mu, \sigma^2)$, $E(X) = \mu$, $Var(X) = \sigma^2$



$P(a \leq X \leq b) = \int_a^b f(x) dx$ (Area under graph)

* Area under graph from $-\infty$ to ∞ must be 1.

* mean, median, mode = μ

$P(X > \mu + k) = P(X < \mu - k)$

$P(|X| < k) = P(-k < X < k)$

$P(X = k) = 0 \Rightarrow P(X \leq k) = P(X < k)$

$P(X \geq k) = P(X > k)$

$P(a \leq X \leq b) = P(X \leq b) - P(X \leq a)$

$P(|X| > k) = P(X > k) + P(X < -k)$

If $X \sim N(\mu, \sigma^2)$,

$nX \sim N(n\mu, n^2\sigma^2)$

$X_1 + X_2 + \dots + X_n \sim N(n\mu, n\sigma^2)$

Standard normal distribution :

$Z = \frac{X - \mu}{\sigma}$

$E(Z) = 0$

$Var(Z) = 1$

$Z \sim N(0, 1)$

$\text{normalcdf}() \rightarrow P(a \leq X \leq b) = ?$

$\text{invnorm}() \rightarrow P(X > \text{or } < ?) = k$

Statistics VI - Sampling and Estimation

Population : Group of items which we want to study.

Parameter : Characteristic of population.

* Defined by Greek letters ($\theta, \mu, \sigma \dots$)

Sample : Finite subset of a population.

Statistics : Characteristic of sample.

* Defined by Roman letters ($T, \bar{X}, s$)

* Sample is used to study characteristics of the population.

* To represent the population properly, sample should be random

Random : Each member of the population has equal chance of being selected.

To create a random sample,

- Create a ~~complete~~ complete list
- Assign each member a unique number
- Generate random number

Estimator : Statistic derived from sample to estimate unknown parameter.

$$\text{Unbiased estimate of } \mu = \bar{X} = \frac{\sum X}{n} = \frac{\sum (X - k) + k}{n}$$

$$\text{Unbiased estimate of } \sigma^2 = s^2 = \frac{n}{n-1} \sigma_{\text{sample}}^2$$

$$= \frac{1}{n-1} \sum (X - \bar{X})^2$$

$$= \frac{1}{n-1} \left[\sum X^2 - \frac{(\sum X)^2}{n} \right]$$

$$= \frac{n}{n-1} \left(\frac{\sum X^2}{n} - \bar{X}^2 \right)$$

$$= \frac{1}{n-1} \left[\sum (X - k)^2 - \frac{(\sum (X - k))^2}{n} \right]$$

Central Limit Theorem : If sample size n is sufficiently large, regardless of distribution,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

OR
 $N\left(\mu, \frac{s^2}{n}\right)$

$$\sum_{i=1}^n X_i \sim (n\mu, n\sigma^2) \text{ APPROXIMATELY}$$

Statistics VII - Hypothesis Testing

Null hypothesis H_0 : Main hypothesis to be tested.

$$\mu = \mu_0$$

Alternative hypothesis H_1 : Another hypothesis which goes against the null hypothesis.

$$\mu > \mu_0$$

OR

$$\mu \neq \mu_0$$

OR

$$\mu < \mu_0$$

Test statistic Z : Random variable derived from sample data.

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \quad \text{OR} \quad \frac{\bar{X} - \mu_0}{s / \sqrt{n}}$$

Critical region : Set of values of Z for which H_0 is rejected.

* Do not reject $H_0 \neq$ Accept H_0

→ Depends on sample

→ Chance for error

Level of significance α : Probability that H_0 is wrongly rejected when H_0 is in fact true.

$$P(H_0 \text{ is rejected} | H_0 \text{ is true})$$

p-value : Smallest value of the significance level α for which H_0 is rejected.

$$H_1 : \mu > \mu_0 \quad | \quad \text{p-value} = P(Z \geq Z_{\text{cal}})$$

$$H_1 : \mu \neq \mu_0 \quad | \quad \text{p-value} = 2P(Z \geq |Z_{\text{cal}}|)$$

$$H_1 : \mu < \mu_0 \quad | \quad \text{p-value} = P(Z \leq Z_{\text{cal}})$$

Test process :

Let μ be ...

Test $H_0 : \mu = \mu_0$

against $H_1 : \begin{cases} \mu > \mu_0 \\ \mu < \mu_0 \end{cases} \quad \text{1-tailed test}$

$\mu \neq \mu_0$ - 2-tailed test

Perform a 1-tailed / 2-tailed test at the $\alpha\%$ significance level.

Under H_0 ,

$\bar{X} \sim N(\mu_0, \frac{\sigma^2}{n})$ OR $\bar{X} \sim N(\mu_0, \frac{s^2}{n})$ APPROXIMATELY
by CLT
since n is large

$$\Rightarrow Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \quad \text{OR} \quad \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim N(0,1)$$

Using GC,

p-value = ...

Rejecting H_0 :

Since p-value $< \alpha$, reject H_0 and conclude that at the $100\alpha\%$ significance level, there is sufficient evidence that ...

No rejecting H_0 :

Since p-value $> \alpha$, do not reject H_0 and conclude that at the $100\alpha\%$ significance level, there is insufficient evidence that ...